Zohaib Aftab, Ph.D.

AI Architecture Specialist | Generative AI & Enterprise Solutions
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Summary

AI architect and data scientist with 10+ years of experience across industry and research. Designs and deploys enterprise AI, machine learning, and generative AI solutions on Azure. Combines hands-on engineering with reference architecture design, cross-functional technical leadership, and MLOps/LLMOps practices. Ph.D. in robotics; bilingual in French and English.

Experience

Sept 2024 – present

Specialist - AI
Synechron

AI solution architecture and delivery for US Bank wealth and asset management.

- Led a production generative AI assistant integrating SQL, Swagger API references, Confluence, and SharePoint through retrieval-augmented generation (RAG). Adopted by 600 users, it reduced API discovery time by 90% and saved over $1M in annual operating costs.
- Designed Python microservices with async FastAPI, SQLAlchemy, and connection pooling; deployed on Azure Kubernetes Service (AKS) with horizontal pod autoscaling.
- Architected a ServiceNow incident resolution platform with defined service boundaries, reusable interfaces, and validation schemas, enabling engineering teams to integrate their data sources.
- Delivered LangGraph multi-agent workflows across Salesforce, SharePoint, and Confluence. Combined hybrid search with source-grounded answers and evaluated accuracy, groundedness, recall, and latency.
- Architected a document intelligence platform with Azure Document Intelligence and LLMs, using configuration-driven extraction schemas to support new business lines without code changes.
- Establish reference architectures and testing playbooks, mentor engineers, and support pre-sales solution design and technical proposals. Responsibilities include system design and data governance.

Oct 2023 – Sept 2024

Senior Data Scientist
B-yond – Montreal, Canada

- Built a RAG application over 50,000+ pages of technical documentation and PostgreSQL data, achieving over 90% precision and recall.
- Engineered Apache Spark and Databricks pipelines on Azure; developed time-series models for Wi-Fi quality-of-service trends and LDA/BERTopic models for Orange's JIRA issue resolution.
- Collaborated with architecture and frontend teams to deploy AI applications, using CI/CD pipelines and MLflow model tracking for reliable delivery.

July 2018 – June 2023 **Research Data Scientist**
National Center of Robotics and Automation – Pakistan

- Directed cross-functional teams on government projects with budgets exceeding $200,000, delivering 2 commercial AI products and securing 3 patents.
- Built pipelines for 8 million+ data points spanning 11+ hours of sequential data; led temporal CNN development in TensorFlow for human activity recognition.

Technical skills

AI architecture & integration: Solution design, reference architectures, enterprise integration, Python microservices, data governance, SDLC, Scrum

Machine learning & generative AI: Deep learning (CNNs), LLMs, RAG, multi-agent systems, LangGraph, LangChain, MCP, TensorFlow, scikit-learn, Azure Document Intelligence

Cloud & data platforms: Azure, AKS, Docker, Kubernetes, FastAPI, SQLAlchemy, SQL, PostgreSQL, Apache Spark, Databricks

MLOps / LLMOps & observability: CI/CD, GitHub Actions, GitLab pipelines, MLflow, evaluation pipelines, Langfuse, LangSmith, Lumenova

Certifications

2026 **Azure AI Apps and Agents Developer Associate**
Microsoft | Proctored exam (AI-103)

2026 **Claude Certified Architect - Foundations**
Anthropic | Proctored exam (CCA-F)

Education

2012 **Université de Lyon**, Ph.D. in Robotics – France

- Humanoid balance recovery using constrained optimization, quadratic programming, and model predictive control.

2009 **École Centrale Nantes**, Master's in Automation – France

Mentoring & writing

Teach in Johns Hopkins University's Certificate Program in Agentic AI, covering generative AI foundations, single-agent design, multi-agent systems, evaluation, guardrails, and human oversight.

Data Science Mentor at Great Learning since 2024. Mentor experienced professionals in UT Austin's PGP in Data Science and Business Analytics, covering Python, data analysis, supervised and unsupervised learning, ensemble methods, model tuning, and deployment.

Conducted over 200 hours of mentoring and maintained an overall rating of 4.8 out of 5.

Write Scalable ML Stack, a Substack newsletter sharing insights on enterprise-grade software applications.